

CSCI 7350 Data mining

Homework 2

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Learning Objectives

By completing this assignment, you should be able to:

- Understand binary and multiclass logistic regression (Softmax)
- Interpret effects of L1 and L2 regularization
- Compute and interpret evaluation metrics (ROC, PR, F1, AUC)
- Apply models to real datasets and analyze results

Part 1: Conceptual Understanding (Logistic Regression & Softmax)

Q1. Binary Logistic Regression

Given: $P(y = 1 \mid x) = \sigma(w_0 + w_1 x)$, where $w_0 = -2$, $w_1 = 1$

1. Compute $P(y = 1 \mid x)$ for $x = 1, 2, 3$

A scanned copy of the answer has been added at the end of the document.

2. Using threshold = 0.5, determine the predicted class for each.

A scanned copy of the answer has been added at the end of the document.

3. How does increasing w affect the decision boundary?

A scanned copy of the answer has been added at the end of the document.

Q2. Softmax (Multiclass Logistic Regression)

Given logits: $[2.0, 1.0, 0.1]$ for classes A, B, C.

1. Compute Softmax probabilities.

Softmax converts logits $[2.0, 1.0, 0.1]$ into probabilities using

$$P(y_i) = \frac{e^{z_i}}{\sum_j e^{z_j}}$$

$$e^{2.0} \approx 7.389$$

$$e^{1.0} \approx 2.718$$

$$e^{0.1} \approx 1.105$$

So the denominator is $7.389 + 2.718 + 1.105 = 11.212$

Class A: $P(A) = 7.389/11.212 = 0.659$

Class B: $P(B) = 2.718/11.212 = 0.242$

Class C : $P(C) = 1.105/11.212 = 0.099$

2. Which class is predicted?

The predicted class is the one with the highest probability. So, Class A is predicted because it has the highest probability.

3. For class A, explain how evaluation becomes a **One-vs-rest problem**.

In multiclass classification with softmax (classes A, B, C), evaluating a single class can be done using the One-vs-rest approach.

For Class A, we convert the multi-class problem into a binary classification problem.

Positive class: A

Negative class: all other classes (B and C)

So, with this, here is the following table

Actual Class	Binary Level for Class A
A	1(positive)
B	0 (rest)
C	0 (rest)

So, if the model predicts A, the prediction is positive

If the model predicts B or C, the prediction is negative.

4. If all logits are divided by 2 (temperature-like smoothing), how do probabilities change qualitatively?

If the logits are divided by 2, the softmax will be

$$P(y_i) = e^{z_i/2} / \sum_j e^{z_j/2}$$

So the difference between class probabilities becomes smaller. So, our previously predicted Class A probability will be lower than before. On the contrary, Classes B and C have a higher probability than before.

Overall, larger logits imply a very confident prediction, and smaller logits suggest softer probabilities.

Part 2: Evaluation Metrics

Q3. Confusion Matrix & Metrics

Given:

TP	FP	TN	FN
40	10	30	20

Compute:

- Precision

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP}) = 40 / (40 + 10) = 40 / 50 = 0.80$$

- Recall

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN}) = 40 / (40 + 20) = 40 / 60 = 0.667$$

- F1 score

$$\text{F1} = 2 * \text{Precision} * \text{Recall} / (\text{Precision} + \text{Recall}) = 2 * 0.8 * 0.667 / (0.8 + 0.667) = 1.067 / 1.467 = 0.727$$

- Accuracy

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN}) = (40 + 30) / (40 + 30 + 10 + 20) = 70 / 100 = 0.70$$

Q4. ROC Curve Construction

Given:

Score Label

0.9 1

0.8 0

0.7 1

0.6 0

1. Sort by score and simulate threshold scanning.

Row with label 1 is (0.9, 1) and (0.7, 1). So there are two positives. So, $P=2$

Row with label 0 is (0.8, 0) and (0.6, 0). So, there are two negatives as well. So, $N=2$

$$TPR = TP / P$$

$$FPR = FP / N$$

2. Compute TPR and FPR at each step.

We move the threshold from the **highest score to the lowest**.

Step 1 — Threshold > 0.9

No samples predicted positive.

TP	FP	FN	TN
----	----	----	----

0	0	2	2
---	---	---	---

$$TPR=0/2=0 \quad TPR = 0/2 = 0 \quad TPR=0/2=0 \quad FPR=0/2=0 \quad FPR = 0/2 = 0 \quad FPR=0/2=0$$

Point $\rightarrow (0,0)$

Step 2 — Threshold = 0.9

Predicted positive: **0.9**

TP	FP	FN	TN
----	----	----	----

1	0	1	2
---	---	---	---

$$TPR=1/2=0.5 \quad TPR = 1/2 = 0.5 \quad TPR=1/2=0.5 \quad FPR=0/2=0 \quad FPR = 0/2 = 0 \quad FPR=0/2=0$$

Point $\rightarrow (0, 0.5)$

Step 3 — Threshold = 0.8

Predicted positive: **0.9, 0.8**

TP	FP	FN	TN
----	----	----	----

1	1	1	1
---	---	---	---

$TPR=1/2=0.5$ $TPR = 1/2 = 0.5$ $TPR=1/2=0.5$ $FPR=1/2=0.5$ $FPR = 1/2 = 0.5$ $FPR=1/2=0.5$

Point $\rightarrow$ **(0.5, 0.5)**

Step 4 — Threshold = 0.7

Predicted positive: **0.9, 0.8, 0.7**

TP	FP	FN	TN
----	----	----	----

2	1	0	1
---	---	---	---

$TPR=2/2=1$ $TPR = 2/2 = 1$ $TPR=2/2=1$ $FPR=1/2=0.5$ $FPR = 1/2 = 0.5$ $FPR=1/2=0.5$

Point $\rightarrow$ **(0.5, 1)**

Step 5 — Threshold = 0.6

Predicted positive: **all samples**

TP	FP	FN	TN
----	----	----	----

2	2	0	0
---	---	---	---

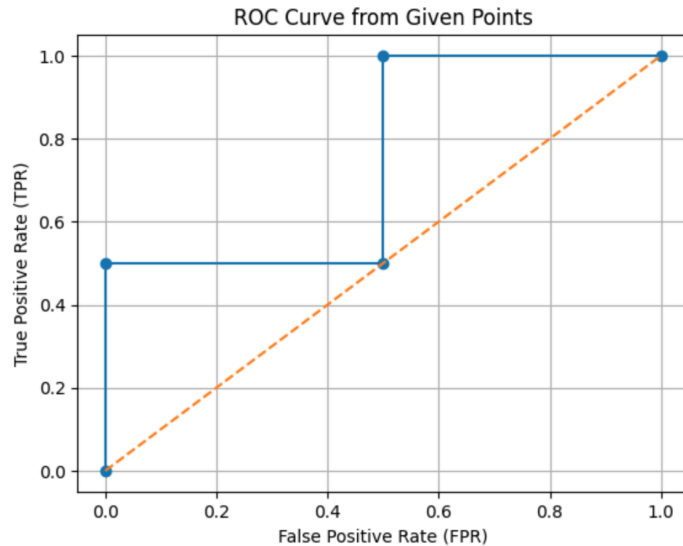
$TPR=2/2=1$ $TPR = 2/2 = 1$ $TPR=2/2=1$ $FPR=2/2=1$ $FPR = 2/2 = 1$ $FPR=2/2=1$

Point $\rightarrow$ **(1,1)**

3. Sketch the ROC curve.

ROC points:

$(0,0) \rightarrow (0,0.5) \rightarrow (0.5,0.5) \rightarrow (0.5,1) \rightarrow (1,1)$



Overall, the curve is above the diagonal line, indicating classifier is performing better than random guessing.

4. Explain the meaning of AUC in this context.

AUC (Area Under the Curve) measures the model's ability to **separate positive and negative classes**. For the perfect classifier, **AUC = 1**, AUC=0.5 suggests random guessing, and AUC within the range of 0.5 to 1 indicates a useful classifier.

In this dataset, the model ranks most positive examples higher than negatives, so the ROC curve lies above the diagonal, indicating good discrimination.

Q5. Multiclass Metrics (Macro vs Micro)

Given:

True: A, C, A, A, B, A

Predicted: A, B, C, A, A, A

1. Compute precision and recall for each class.

True labels:

A, C, A, A, B, A

Predicted labels:

A, B, C, A, A, A

Classes: **A, B, C**

Total samples = **6**

Class A

Actual A = 4

Predicted A = 4

- **TP(A)** = 3 (positions 1,4,6)
- **FP(A)** = 1 (position 5)
- **FN(A)** = 1 (position 3)

Precision(A)

$$TP / (TP + FP) = 3 / (3 + 1) = 3/4 = 0.75$$

Recall(A)

$$TP / (TP + FN) = 3 / (3 + 1) = 3 / 4 = 0.75$$

Class B

Actual B = 1

Predicted B = 1

- **TP(B)** = 0
- **FP(B)** = 1 (position 2)
- **FN(B)** = 1 (position 5)

Precision(B)

$$0 / (0 + 1) = 0$$

Recall(B)

$$0 / (0 + 1) = 0$$

Class C

Actual C = 1

Predicted C = 1

- **TP(C)** = 0

- **FP(C)** = 1 (position 3)
- **FN(C)** = 1 (position 2)

Precision(C)

$$0 / (0 + 1) = 0$$

Recall(C)

$$0 / (0 + 1) = 0$$

Class	Precision	Recall
-------	-----------	--------

A	0.75	0.75
---	------	------

B	0	0
---	---	---

C	0	0
---	---	---

2. Compute:

- **Macro precision**

Macro precision = average of precision across classes = $0.75 + 0 + 0 / (3) = 0.25$

Macro Precision = 0.25

- **Micro precision**

First sum global TP and FP.

TP total

$$3 + 0 + 0 = 3 \quad 3 + 0 + 0 = 3$$

FP total

$$1+1+1=3 \quad 1 + 1 + 1 = 3 \quad 1+1+1=3$$

Micro precision

$$3 / (3+3) = 0.5$$

Micro Precision = 0.5

Explain the difference in interpretation between macro and micro.

Macro precision, compute precision for each class separately, then average them. Because it treats all classes equally, it is useful when class distribution is imbalanced, and we want fairness across classes.

Micro Precision computes metrics globally across all samples. It uses total TP and FP. This method weights classes by their frequency.

Part 3: Real Data Analysis – Binary Classification

Q6. Logistic Regression + Evaluation

Use Breast Cancer data set (binary class)

Tasks

- Train logistic regression models:
 - Without regularization
 - With L2 regularization (try multiple values)
- Evaluate using:
 - Accuracy, Precision, Recall, F1
 - ROC curve + AUROC

- PR curve + AUPRC

Questions:

1. Which model performs best? Explain using metrics.

Comparison of Performance Metrics

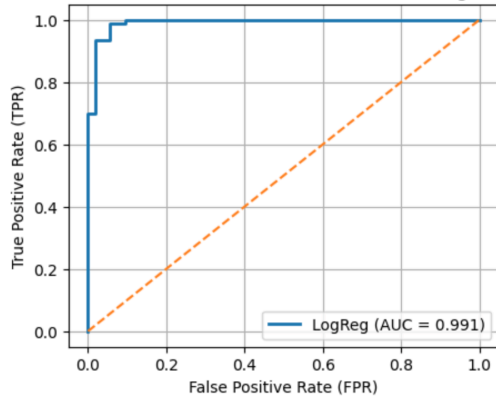
Metric	Without Regularization	With L2 Regularization
Accuracy	0.96	0.98
Precision (Malignant)	0.94	0.98
Recall (Malignant)	0.94	0.98
F1-score (Malignant)	0.94	0.98
Precision (Benign)	0.97	0.99
Recall (Benign)	0.97	0.99
F1-score (Benign)	0.97	0.99

The **regularized model performs better based on all key metrics**. As we observe from the above table, the overall **accuracy increases from 0.96 to 0.98**, indicating that the regularized model correctly classifies a larger proportion of samples. For the **malignant class**, which is particularly important in medical diagnosis, the **precision, recall, and F1-score increase from 0.94 to 0.98**. This means the model makes fewer false positives and fewer false negatives when identifying malignant tumors. Similarly, the **benign class metrics improve from about 0.97 to 0.99**, showing better overall classification reliability.

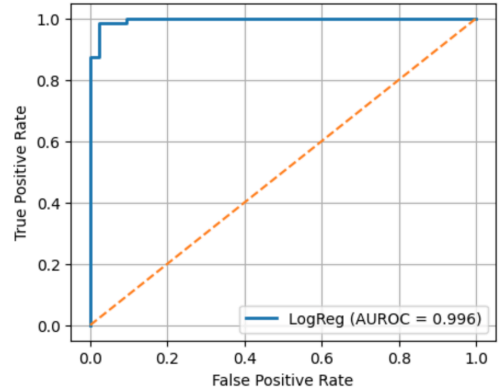
According to our results, we find that regularization helps by **penalizing large model coefficients**, which reduces overfitting and improves the model's ability to generalize to unseen test data. As a result, the logistic regression model with **L2 regularization (C = 0.1)** yields the best performance considering the accuracy, precision, recall, and F1-score, and is therefore the preferred model for this dataset.

2. Compare ROC and PR curves. Do they tell the same story?

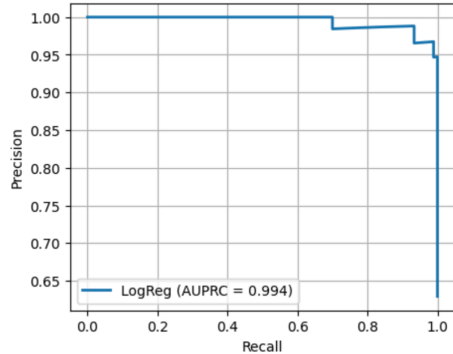
ROC Curve — Breast Cancer (Sklearn) without Regularization



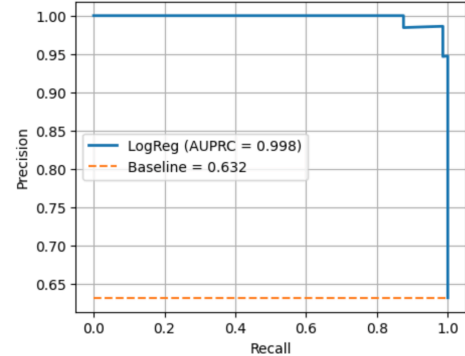
ROC Curve — Breast Cancer with L2 regularization



Precision-Recall Curve — Breast Cancer (Sklearn) without regularization



Precision-Recall Curve — Breast Cancer with L2 regularization



Yes, based on our results, the ROC curves tell **the same qualitative story**: both models perform extremely well. However, the **logistic regression model with L2 regularization performs slightly better**, as we get a higher AUROC and slightly improved curve position. Regularization likely improves **generalization and reduces overfitting**, resulting in this small performance gain.

The PR curves confirm the results seen in the ROC curves and classification metrics. **Both models perform extremely well**, but the **logistic regression model with L2 regularization slightly outperforms the non-regularized model**, as shown by the higher AUPRC (0.998 vs 0.994) and the slightly higher precision across recall values.

3. How does regularization affect model performance?

The softmax model has many parameters. For the high-dimensional data, it has risk of overfitting, unstable prediction, and large coefficients. That's why we need regularization in softmax. Regularization changes learning.

As we discussed in the last question, regularization has a **significant impact on model performance** by controlling model complexity and improving generalization. In logistic regression with **L2 regularization**, the parameter **C** is in inverse relation with the regularization strength, which means a small C value means stronger regularization and a large C value means weak regularization.

The model performance with c values is shown below:

C	Accuracy	Precision	Recall	F1	ROC_AUC	AUPRC
0.001	0.938596	0.957746	0.944444	0.951049	0.990741	0.994654
0.010	0.956140	0.958904	0.972222	0.965517	0.995040	0.997140
0.100	0.982456	0.986111	0.986111	0.986111	0.996032	0.997645
1.000	0.982456	0.986111	0.986111	0.986111	0.995701	0.997297
10.000	0.964912	0.972222	0.972222	0.972222	0.993386	0.995726
100.000	0.947368	0.971429	0.944444	0.957746	0.986111	0.990788

Regularization boosts model performance by preventing overly large coefficients and reducing overfitting. As shown in the above tabular form, the **moderate regularization (C =0.1 or C=1)** gives the best performance, achieving the highest accuracy, F1-score, and AUC metrics. Too much regularization causes underfitting, while too little regularization slightly degrades generalization performance.

Part 4: Multiclass Classification (Softmax)

Use wine dataset for this part.

Tasks

- Train a Softmax model
- Evaluate using:
 - Accuracy
 - Macro F1
 - Micro F1
- Plot: ROC curves (per class), micro-average ROC curve and macro-average ROC curve

Questions:

1. Which class is hardest to classify? How can you tell?

Predicted classes: [0 1 0 1 1 0 0 1 1 2]

Class probabilities (sample 1):
[9.99671e-01 3.07000e-04 2.30000e-05]

Class probabilities (sample 2):
[0.007051 0.540423 0.452526]

Class probabilities (sample 3):
[9.920e-01 7.664e-03 3.360e-04]

Class probabilities (sample 4):
[0.289967 0.707467 0.002565]

Class probabilities (sample 5):
[0.033032 0.962313 0.004655]

Class probabilities (sample 6):
[9.99166e-01 7.70000e-04 6.30000e-05]

As we observe in the above results for sample 2, Class **2** tends to be the hardest because its feature distribution overlaps more with **Class 1**, leading to lower confidence predictions and occasional misclassification.

2. Compare macro vs micro metrics. What do they reveal?

The **Micro F1 score** aggregates the contributions of all classes by counting the total **true positives, false positives, and false negatives** across the dataset. In multiclass classification, the **Micro F1 score is usually equal to accuracy**, which is why both values are **0.9722** in our results.

This indicates that the model performs **very well overall**, correctly classifying about **97.22% of the samples**.

The **Macro F1 score** calculates the **F1 score for each class individually and then averages them**. Each class is treated **equally**, regardless of how many samples belong to that class.

We find the Macro F1 score is **0.971**, which is extremely close to the Micro F1 score.

The close agreement between **Macro F1 and Micro F1** indicates that the classifier performs **uniformly across all classes**. There is no major bias toward or against any specific class, and the model achieves **high and balanced performance on the multiclass Wine dataset**.

Part 5: Experimental Procedure & Model Evaluation

Q7. Train/Test Split vs Cross Validation

1. Explain the difference between:
o a single train/test split
o k-fold cross-validation

A **single train/test split is less computationally expensive but less reliable**, while **k-fold cross-validation provides a more accurate evaluation** because it tests the model on multiple different subsets of the data. In our case, both methods show **very high performance (~97–100%)**, confirming that the Softmax logistic regression model performs very well on the Wine dataset.

2. Why might a single split give misleading results?

A **single train/test split** may give misleading results because it **test the model on only one subset of data**, which may not represent the full data distribution. Methods like **k-fold cross-validation** yield **more reliable estimates** by testing the model on multiple different splits and averaging the results.

3. In k-fold cross-validation:

- o what is averaged?

In the K-fold cross-validation, the dataset is split into **k parts (folds)**. The model is trained on these different **k parts**. A performance metric is computed **corresponding to each fold or each part of the training**. Finally, the **metrics from all folds/parts are averaged**.

- o why does this give a more reliable estimate?

Averaging gives a **more reliable estimate of model performance** because it reduces the effect of **one lucky or unlucky split**. It shows **robustness of the model's performance**.

4. When would you prefer:

- o cross-validation
- o a fixed train/test split

When we have millions of samples, a single, large train/test split is sufficient and computationally cheaper than re-training a model k-times.

We prefer cross-validation when dealing with small, limited data, and we want to see the robustness of the model. Cross-validation for the large dataset can be computationally expensive.

Q8. k-Fold Cross Validation

Using the dataset from Part 3 (Breast Cancer):

1. **Perform 5-fold cross-validation** for logistic regression.
I have performed a 5-fold cross-validation as shown in the notebook
2. **Report: mean accuracy (standard deviation)**
I find the mean accuracy from the breast cancer dataset as 0.9737

3. Compare CV result vs your previous single test split result, which one do you trust more and why

For the Breast Cancer dataset, the **single train/test split** produced an **accuracy of 0.9580**, while the **5-fold cross-validation** resulted in a **mean accuracy of 0.9737**.

There is a difference in the results because a single train/test split evaluates the model using only one particular partition of the data. If the chosen test set happens to contain more difficult samples or an unrepresentative distribution of cases, the measured accuracy may be lower or higher than the model's true performance.

In contrast, **5-fold cross-validation** divides the dataset into five subsets and trains the model on five different parts, each time using a different fold as the test set. The performance is then averaged across all folds. Because every observation is used for both training and testing at different stages, cross-validation provides a **more stable and reliable estimate of the model's generalization performance**. In this case, the higher mean accuracy from cross-validation suggests that the model generally performs slightly better than what was indicated by the single train/test split.

Q9. Learning Curve (Using logistic regression from Part 3:

1. What is a learning curve?
 - What is plotted on the x-axis? And the y-axis?

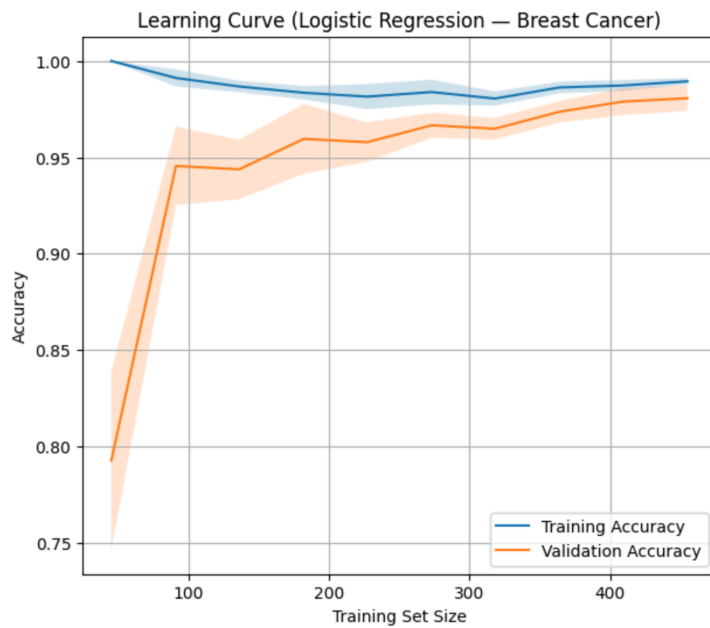
A **learning curve** is a graph that shows how a machine learning model's performance changes as the **amount of training data increases**. It helps to evaluate how well the model learns from data and whether the model is **underfitting, overfitting, or generalizing well**.

On a learning curve, the **x-axis represents the training set size**, which is the number (or proportion) of training examples used to train the model. As we move to the right, the model is trained with more data.

The **y-axis represents the model performance**, typically measured using a metric such as **accuracy, error rate, or F1-score**.

2. Generate a learning curve using cross-validation for breast cancer data using logistic regression.

We have generated the learning curve as shown in the notebook.



3. Based on your plot, compare training and validation curves, determine whether the model is:

- o Overfitting,
- o Underfitting,
- or o Well-fitted.

Explain your answer.

Training Accuracy (Blue Curve)

- The **training accuracy starts very high (~1.0)** when the training set is small.
- As the training set size increases, the training accuracy **slightly decreases to around 0.99**.
- This happens because with more training data, the model cannot memorize every example and must **generalize better**.

Validation Accuracy (Orange Curve)

- The **validation accuracy starts lower (~0.79)** when the training data is very small.
- As more training data is added, the validation accuracy **increases steadily**.
- Eventually it **stabilizes around ~0.98**, showing that the model improves with more data.

Gap Between Training and Validation Curves

- At small training sizes, there is a **large gap** between training and validation accuracy.
- This indicates **overfitting**, because the model fits the small training set very well but performs worse on unseen data.
- As the training size grows, the **gap becomes small**, meaning the model generalizes well.

Convergence of the Curves

- Toward the right side of the plot, the curves **almost meet**.
- Training accuracy ≈ 0.99
- Validation accuracy ≈ 0.98

Variance Bands (Shaded Areas)

The shaded regions represent **standard deviation across folds** during cross-validation.

- Narrow bands mean **consistent performance**
- The model is **stable across different training splits**

This learning curve shows that **logistic regression performs very well on the Breast Cancer dataset**. As more training data is used, the validation accuracy improves and stabilizes close to the training accuracy, indicating **good generalization with minimal overfitting**. Adding more data may provide **only small improvements**, since the model is already near its optimal performance.

Ans 1: The logistic regression probability is

$$P(y=1|x) = \sigma(w_0 + w_1 x)$$

where the Sigmoid function is

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

Given $w_0 = -2$, $w_1 = 1$

$$\begin{aligned} \text{So } P(y=1|x) &= \frac{1}{1 + e^{-(-2+x)}} \\ &= \frac{1}{1 + e^{2-x}} \end{aligned}$$

1. When $x=1$

$$z = -2 + 1 = -1$$

$$P(y=1|x=1) = \frac{1}{1 + e^{-(-1)}} = \frac{1}{1 + e^1}$$

$$e^1 \approx 2.718$$

$$P(y=1|x=1) = \frac{1}{1 + 2.718} \approx 0.269$$

2. When $x=2$,

$$z = -2 + 2 = 0$$

$$P(y=1|x=2) = \frac{1}{1+e^0} = \frac{1}{2} = 0.5$$

3. When $x=3$,

$$z = -2 + 3 = 1$$

$$P(y=1|x=3) = \frac{1}{1+e^{-1}}$$

$$e^{-1} \approx 0.368$$

$$P(y=1|x=3) = \frac{1}{1+0.368} = \frac{1}{1.368} \approx 0.731$$

Q1.2

Using Threshold = 0.5, the prediction rule in logistic regression is

$$\hat{y} = \begin{cases} 1 & \text{if } P(y=1|x) \geq 0.5 \\ 0 & \text{if } P(y=1|x) < 0.5 \end{cases}$$

From the previous calculation,

x	$z = w_0 + w_1 x$	$P(y=1 x)$	Predicted class
1	-1	0.269	0
2	0	0.50	1
3	1	0.731	1

Q1.3 For logistic regression

$$P(y=1|x) = \sigma(w_0 + w_1 x)$$

the decision boundary occurs when the probability is 0.5

Since $\sigma(z) = 0.5$ when $z=0$

the boundary is determined by

$$w_0 + w_1 x = 0$$

$$\Rightarrow x = -\frac{w_0}{w_1}$$

we expect following with increasing w_1

1. The location decision boundary changes because $x = -\frac{w_0}{w_1}$

2. If w_1 increases (while w_0 stays fixed), the value of x decreases, so the boundary shifts toward smaller x values.

3. The sigmoid curve also becomes steeper, making the probability changes more sharply near the boundary.

overall, increasing w_1 shifts the decision boundary and makes the model transition ~~more~~ sharper.